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Adjusting fluid injection into a wellbore based on relative permeability of a formation

Abstract

A fluid injection process for injecting fluid into a wellbore can be adjusted based on relative permeability. An actual trapped gas saturation of a formation can be determined from a maximum gas saturation, a maximum trapped gas saturation, and an actual gas saturation. A pseudo-maximum gas saturation can be determined from the actual trapped gas saturation, the maximum trapped gas saturation, and actual gas saturation. A relative permeability of the formation can be determined by mapping the pseudo-maximum gas saturation along a drainage curve. The fluid injection process can be adjusted based on the relative permeability.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5027898	12/1990	Naae	166/305.1	E21B 43/24
5528030	12/1995	Mickael	250/269.4	G01V 5/107
2002/0173915	12/2001	Egermann	702/12	G01N 15/08
2008/0154563	12/2007	Kumar	703/10	E21B 43/00
2015/0142407	12/2014	Wakefield	703/10	G01V 1/28
2015/0346010	12/2014	Matringe et al.	N/A	N/A
2016/0177686	12/2015	Carvajal et al.	N/A	N/A
2020/0202060	12/2019	Thakur	N/A	E21B 43/16
2021/0381363	12/2020	Hadibeik et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111814347	12/2019	CN	E21B 47/00
2007076044	12/2006	WO	N/A

OTHER PUBLICATIONS

Land, “Calculation of Imbibition Relative Permeability for Two- and Three-Phase Flow From Rock Properties”, SPE Journal 8, Jun. 1968, pp. 149-156. cited by applicant

Larsen et al., “Methodology for Numerical Simulation With Cycle-Dependent Relative Permeabilities”, SPE Journal Jun. 1998, pp. 163-173. cited by applicant

International Patent Application No. PCT/US2022/082654, International Search Report and Written Opinion mailed Sep. 20, 2023, 8 pages. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to hydrocarbon recovery operations and, more particularly (although not necessarily exclusively), to fluid-injection operations for a wellbore.

BACKGROUND

(2) Water-alternating-gas injection may be a technique employed to extract hydrocarbons from a subsurface formation that may not otherwise be extractable. The technique may involve alternatively injecting a water-based solution and carbon dioxide into a wellbore to displace

hydrocarbons more effectively within a subsurface formation. The state of matter, or phase, of fluids injected into the wellbore can change unpredictably. These phase changes can affect the recoverability of hydrocarbons in the formation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic of a production and fluid injection system for a subsurface formation according to some aspects of the present disclosure.
- (2) FIG. 2 is a block diagram of a computing device for modelling a fluid injection process according to some aspects of the present disclosure.
- (3) FIG. 3 is a flowchart for a process of adjusting a fluid injection into a formation based on a relative permeability of the formation according to some aspects of the present disclosure.
- (4) FIG. 4 is a graph of curves modelling gas saturation data against a relative permeability according to some aspects of the present disclosure.
- (5) FIG. 5 is a phase diagram of a computer model for fluid injected into a subsurface formation according to some aspects of the present disclosure.

DETAILED DESCRIPTION

- (6) Certain aspects and examples of the present disclosure relate to adjusting fluid injection into a wellbore based on a relative permeability of a subsurface formation. Relative permeability can be a ratio of the effective permeability of a particular fluid at a particular saturation against the absolute permeability of the same fluid at total saturation. Effective permeability of a subsurface formation can be a measure of the ability of that particular fluid to flow in the presence of other fluid phases. Saturation may be the relative amount of water, carbon dioxide, and hydrocarbon in the pores of the subsurface formation.
- (7) The relative permeability of a subsurface formation can provide an estimate of total hydrocarbon reserves within the subsurface formation. As a result, relative permeability can be used to inform decisions about rates of hydrocarbon production from the subsurface formation. Hydrocarbon production from the subsurface formation, in turn, can be regulated by adjusting fluid injection into a wellbore of the subsurface formation.
- (8) Data for relative permeability to a fluid can be determined based on gas saturation data. But, it may be challenging to determine relative permeability data when the value for gas saturation data falls below the value for trapped gas saturation data. Trapped gas saturation data may be a saturation data value at which a fluid mixture from a water-alternating-gas hydrocarbon recovery process may have no movement within pores of a subsurface formation unless gas saturation is increased.
- (9) In some examples of the present disclosure, a relative permeability of a subsurface formation can be determined with a given water and carbon-dioxide based fluid mixture and with a given gas saturation data value, regardless of the gas saturation data value being above or below the trapped gas saturation data value. By determining a relative permeability with a gas saturation data value regardless of the value's quantity relative to a trapped gas saturation data value, the total hydrocarbon reserves of a subsurface formation can be estimated with a wider range of data. With more computationally accessible estimates of total hydrocarbon reserves in a subsurface formation, water-alternating-gas operations can be more effectively controlled so that hydrocarbon production can align with a demand for the hydrocarbons or capacity for hydrocarbon extraction.
- (10) Also, determining a relative permeability, regardless of a saturation value relative to a trapped gas saturation data value, can more accurately estimate a capacity of a depleted subsurface formation for the purpose of a carbon dioxide geological sequestration process. The relative permeability of the depleted subsurface formation may determine a storage rate of carbon dioxide

or inform the design of a surface facility for sequestering carbon dioxide. Also, condensation effects resulting from phase changes to the sequestered carbon dioxide may be estimated by the actual trapped gas saturation data. Removing carbon dioxide from the atmosphere and sequestering the carbon dioxide in depleted subsurface formations can mitigate or defer effects of global warming, which may help avoid dangerous weather events that may be caused by climate change. (11) In addition to determining relative permeability regardless of a saturation value relative to a trapped gas saturation data value, the relative permeability can be determined at a maximum gas saturation data value without requiring the model effects of interfacial tension. Interfacial tension may be an accumulation of energy and an imbalance of force at an interface of two distinct phases of material, such as a liquid phase and a gaseous phase. By avoiding consideration of interfacial tension, time and resources may be saved by avoiding tests that may be necessary to determine interfacial tension.

(12) Water-alternating-gas production simulations can depend on modeling relative permeability. Subsurface formations in which fluids are near or have reached a supercritical phase may be particularly dependent on modeling. During an actual water-alternating-gas production scenario, trapped gas saturation data may be obtained at the wetting phase. At this point, simulated gas may have no mobility within the subsurface formation unless simulated gas saturation is increased. It may be less clear how to model cases where gas saturation data value drops below the trapped gas saturation data threshold due to phase behavior and then increases due to displacement from fluid injection.

(13) In some examples of a water-alternating-gas production simulation, the simulation may begin with the fluid mixture in a gaseous phase. This may cause a large, implied trapped gas saturation data value. The simulation may proceed to a point where the fluid mixture injected into the formation and hydrocarbons within the formation become a single phase, supercritical oil. Alternatively, the simulation may proceed to a point where the fluid mixture injected into the formation and the hydrocarbons within the formation become subcritical. For the case of the supercritical oil, previously disclosed simulations that may not be able to account for gas saturation data value below a trapped gas saturation data value and may report an unlikely predicted outcome in which some or all of the simulated trapped gas saturation disappears.

(14) In some examples of a water-alternating-gas simulation where relative permeability can be reliably determined regardless of gas saturation data being above or below the trapped gas saturation data, fluid injection can be adjusted with a wider set of simulation inputs. As a result, fluid injection can be better controlled with regard to either the amount of fluid injected into a wellbore, or the composition of the fluid injected into the wellbore. Additionally or alternatively, phases of fluid injection can be better controlled. For example, relative permeability can help determine the volume and duration of a wetting phase, in which a low-salinity water-based solution is injected into the wellbore and a non-wetting phase, in which a carbon-dioxide gas is injected into the wellbore. In some examples, the relative permeability, as an indication of a subsurface formation's total reserves of hydrocarbons, can inform the design of a surface facility intended for extracting the hydrocarbons.

(15) Illustrative examples are given to introduce the reader to the general subject matter discussed herein and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

(16) FIG. 1 is a schematic of a production and fluid injection system **100** for a subsurface formation **114** according to some aspects of the present disclosure. The production wells **106** may allow fluids from the subsurface formation **114** to flow through their completion **108** and to the surface, where production tubing **105** may carry the fluids to production gathering facilities **104**, and in turn, to a separator **102**. The separator **102** system may isolate each fluid phase (typically oil,

gas, and water). In some cases, the water or gas produced are separated and then sent to an injection distribution system **110**. The injection distribution system **110** can also receive injection fluids from exterior sources. The injection wells **112** may receive the fluids to be injected from the injection distribution system **110** via a network of pipelines and inject these fluids in the subsurface formation **114** through well completions **108**.

(17) In some embodiments, the reservoir fluid mixtures may be composed of two or more phases including oil, water, or gas. The subsurface formation **114** itself may be composed of multiple different wells or well blocks. These well blocks may have different physical properties. For example, the well blocks may have formed in different manners geologically. The wells may include different amounts of oil, gas, water, or other materials. Still further, the well blocks may be subject to different pressures owing to the varied materials amounts, or other forces such as the injection of fluids in adjacent or neighboring wells. Moreover, each well in the subsurface formation **114** may have stronger or weaker connections to other wells in the reservoir.

Accordingly, in such cases a computer system may be used to model physical material flow relationships between wells in the subsurface formation **114**, including connections between the injection well **112** and production wells **106** as well as between wells and aquifers.

(18) FIG. 2 is a block diagram of a computing device **200** for modelling a fluid injection process according to some aspects of the present disclosure. The computing device **200** includes a processor **202** communicatively coupled to a memory **204**. The processor **202** can include one processing device or multiple processing devices. Non-limiting examples of the processor **202** include a Field-Programmable Gate Array (FPGA), an application-specific integrated circuit (ASIC), or a microprocessor. The processor **202** can execute instructions **206** stored in the memory **204** to perform computing operations. The instructions **206** may include processor-specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, such as C, C++, C #, Python, or Java.

(19) The memory **204** can include one memory device or multiple memory devices. The memory **204** can be volatile or can be non-volatile, such that it can retain stored information when powered off. Some examples of the memory **204** can include electrically erasable and programmable read-only memory (EEPROM), flash memory, or any other type of non-volatile memory. At least some of the memory **204** includes a non-transitory computer-readable medium from which the processor **202** can read instructions **206**. A computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the processor **202** with computer-readable instructions or other program code. Some examples of a computer-readable medium include magnetic disks, memory chips, ROM, random-access memory (RAM), an ASIC, a configured processor, optical storage, or any other medium from which a computer processor can read the instructions **206**. The memory **204** also includes values for maximum gas saturation data **208**, maximum trapped gas saturation data **210**, actual gas saturation data **212**, actual trapped gas saturation data **214**, pseudo-maximum gas saturation data **216**, and relative permeability data **218** of a subsurface formation.

(20) Maximum trapped gas saturation data **210** and actual gas saturation data **212** may be derived from analysis of rock core plugs from a subsurface formation. In such experiments, the rock core plug may be fully saturated in hydrocarbons. The hydrocarbons may be dispersed by a laboratory treatment approximating a water-alternating-gas hydrocarbon recovery process. The hydrocarbons may be dispersed from the rock core plug until an irreducible saturation point is reached, at which point the fraction of porous space in the rock core plug occupied by fluid from the water-alternating-gas hydrocarbon recovery process is at maximum.

(21) Maximum gas saturation data **208** may be a value useful for determining relative permeability data **218** of a subsurface formation. Maximum gas saturation data **208** may be determined from maximum trapped gas saturation data **210**, actual gas saturation data **212**, and a synthetic value for actual trapped gas saturation data **214**. Actual trapped gas saturation data **214** may be a value useful

for determining relative permeability data **218** of a subsurface formation.

(22) Actual trapped gas saturation data **214** may be determined from maximum trapped gas saturation data **210**, actual gas saturation data **212**, and maximum gas saturation data **208**. For example, (Eq. 1) may be used to determine actual trapped gas saturation data **214**. In such an example, let maximum trapped gas saturation data **210** be S_{gr} , let actual gas saturation data **212** be S_{gu} , let a synthetic value for actual trapped gas saturation data **214** be S_{grap} , and let maximum gas saturation data **208** be S_{gmax} . In such an example, the following equation may be used to determine maximum gas saturation data **208**.

$$(23) \quad S_{grap} = \frac{S_{gmax}}{(S_{gmax}((\frac{1}{S_{gr}}) - (\frac{1}{S_{gu}})) + 1)} \quad (\text{Eq. 1})$$

(24) Pseudo-maximum gas saturation data **216** may be a value useful for determining relative permeability data **218** of a subsurface formation. Additionally or alternatively, pseudo-maximum gas saturation data may be determined for subsequent values of the actual trapped gas saturation data **214** from hysteresis modeling accounting for fluid entrapment, in the formation, from the non-wetting phase when the fluid entrapment from the wetting-phase is increased. Pseudo-maximum gas saturation data **216** may be determined from actual trapped gas saturation data **214**, maximum trapped gas saturation data **210**, and actual gas saturation data **212**.

(25) Relative permeability data **218** can be a ratio of effective permeability of a particular fluid at a particular saturation against the absolute permeability of the same fluid at total saturation. Effective permeability of a subsurface formation to a fluid phase in pores of the subsurface formation can be a measure of the ability of that phase to flow in the presence of other fluid phases. Saturation may be the relative amount of water, carbon dioxide, and hydrocarbon in the pores of the subsurface formation. Relative permeability data **218** can be determined by a curve modeling module **220**. The curve modeling module **220** may determine a non-linear relationship between values for gas saturation data and values for relative permeability data **218**.

(26) FIG. 3 is a flowchart **300** for a process of adjusting a fluid injection into a formation based on a relative permeability data **218** of the formation according to some aspects of the present disclosure. Some examples may include more steps, fewer steps, different steps, or a different combination of steps than is shown in FIG. 3. The steps of FIG. 3 are described below with reference to the components of FIG. 2 described above.

(27) In block **302** the processor **202** may determine an actual trapped gas saturation data **214** of a formation from a maximum gas saturation data **208**, a maximum trapped gas saturation data **210**, and an actual gas saturation data **212**. Maximum trapped gas saturation data **210** and actual gas saturation data **212** may be derived from analysis involving rock core plugs from a subsurface formation. Maximum gas saturation data **208** may be determined from maximum trapped gas saturation data **210**, actual gas saturation data **212**, and a synthetic value for an actual trapped gas saturation data **214**.

(28) In block **304**, the processor **202** may determine a pseudo-maximum gas saturation data **216** from the synthetic value for the actual trapped gas saturation data **214**, the maximum trapped gas saturation data **210**, and the actual gas saturation data **212** using an inverse Land function. For example, let maximum trapped gas saturation data **210** be S_{gr} , let actual gas saturation data **212** be S_{gu} , let the synthetic value for actual trapped gas saturation data **214** be S_{grap} , and let pseudo-maximum gas saturation data **216** be S_{gmax_adj} . In such an example, the following equation may be used to determine pseudo-maximum gas saturation data **216**.

$$(29) \quad S_{gmax_adj} = \frac{1}{((\frac{1}{S_{grap}}) - (\frac{1}{S_{gr}}) + (\frac{1}{S_{gu}}))} \quad (\text{Eq. 2})$$

(30) Before determining the pseudo-maximum gas saturation data **216**, the processor **202** may determine the actual trapped gas saturation data **214** using a Land function, such as (Eq. 1) and the value for actual trapped gas saturation data **214** from block **302**. The processor **202** may determine the pseudo-maximum gas saturation data **216**.

- (31) In block **306** the processor **202** may determine a relative permeability data **218** of the formation by mapping the pseudo-maximum gas saturation data **216** along a drainage curve. A drainage curve may correspond values for relative permeability data **218** with values for gas saturation. A drainage curve may be selected from a plurality of drainage curves based on the value of actual gas saturation data **212** relative to actual trapped gas saturation data **214**.
- (32) In block **308** the processor **202** may adjust fluid injection into a formation based on the relative permeability data **218**. The relative permeability data **218** may indicate a quantity of total hydrocarbon reserves within the formation. The fluid injection may be adjusted to affect a hydrocarbon production rate of the formation. The particular amounts of carbon dioxide gas and water-based solution may also be adjusted, based on the relative permeability data **218**, to affect the hydrocarbon production rate. For example, relative permeability can help determine the volume and duration of a wetting phase, in which a low-salinity water-based solution is injected into the wellbore and a non-wetting phase, in which a carbon-dioxide gas is injected into the wellbore. In such an example, an automatic adjustment of the wetting phase and the non-wetting phase may be controlled by the processor **202** based on the relative permeability data **218**. The hydrocarbon production rate may be adjusted based on considerations such as market demand for hydrocarbons, a capacity for shipping hydrocarbons away from the well site, or a capacity for refining the hydrocarbons away from the well site. Additionally or alternatively, the relative permeability data **218** of a formation may inform a design of a surface facility for extracting hydrocarbons from the formation.
- (33) In some examples, the relative permeability data **218** may indicate a storage capacity of a formation for sequestering carbon dioxide extracted from the atmosphere. Carbon dioxide may be stored in subsurface formations such as depleted oil fields, depleted gas fields, saline formations, saline aquifers, saline-filled basalt formations, and coal seams. Carbon dioxide may be prevented from escaping to a surface by physical mechanisms, such as highly impermeable caprock.
- (34) FIG. **4** is a graph **400** of curves modelling gas saturation data against relative permeability data **404** according to some aspects of the present disclosure. The horizontal axis represents values of gas saturation data **402**. The vertical axis represents values of relative permeability data **404**. A pseudo-maximum gas saturation data **406** value lies along the primary drainage curve **416**. An actual trapped gas saturation data **408** value lies at a point on the graph **400** where relative permeability data **404** is zero. A maximum gas saturation data **410** value also lies along the primary drainage curve **416**.
- (35) A secondary curve **414** between the value for maximum gas saturation data **410** and the value for actual trapped gas saturation data **408** may represent a relationship between values of gas saturation data that may be expressed in (Eq. 1). The secondary curve **414** may allow for limited inferences using the primary drainage curve **416** when gas saturation data values are below the actual trapped gas saturation data **408** value.
- (36) A tertiary curve **412** between the value for pseudo-maximum gas saturation data **406** and a given gas saturation data value may represent a relationship between values of gas saturation data that may be expressed in (Eq. 2). (Eq. 2) and the resulting tertiary curve **412** may be useful for determining relative permeability data **404** values from gas saturation data **402** values when the gas saturation data **402** values are less than the actual trapped gas saturation data **408** value.
- (37) FIG. **5** is a phase diagram **500** of a computer model for fluid injected into a subsurface formation according to some aspects of the present disclosure. The phase diagram **500** may illustrate how a computer model which is not using the techniques described within this disclosure, particularly (Eq. 2), may simulate a phase change that may be physically impossible. The horizontal axis represents temperature **502** and the vertical axis represents pressure **504**. The phase diagram **500** illustrates a gas phase **508** region, an oil phase **506** region, and a critical point **510** at which oil and gas phases may become indistinguishable.
- (38) At a first point **512**, the model may begin simulating a water-alternating-gas production

scenario with the fluid from the injection beginning in a gaseous phase. The model may determine the maximum gas saturation data when simulating the first point **512**. From the first point **512**, the model may simulate a scenario in which the fluid becomes supercritical oil at a second point **514** or a subcritical oil at a third point **516**.

(39) The model may proceed to the supercritical phase at the second point **514** or the subcritical phase at the third point **516** based on the amount and the composition of the fluids injected into the modeled formation, the amount and the composition of the fluids present in the modeled formation, the pressure within the modeled formation, and the temperature within the modeled formation. In an example where the fluid becomes a supercritical oil at the second point **514**, carbon dioxide or water may be dissolved within hydrocarbons from the modeled formation, which may raise the viscosity of a fluid mixture of hydrocarbons, carbon dioxide, and water. The raised viscosity may allow for a more complete recovery of the hydrocarbons from the subsurface formation. In an example where the fluid becomes a subcritical oil at the third point **516**, carbon dioxide and water may not dissolve into hydrocarbons of the modeled formation to sufficiently change the viscosity of the hydrocarbons or improve the recoverability of the hydrocarbons.

(40) In some examples, modeling conditions may cause the simulated supercritical fluid at the second point **514** to fall to a subcritical phase illustrated as a fourth point **518**. This progression, characterized by a fall in pressure **504**, may not be physically possible. The progression from the second point **514** to the fourth point **518** may imply a simulated trapped gas saturation has disappeared without explanation. A model may be improved by utilizing (Eq. 2), which may prevent the model from progressing from the second point **514** to the fourth point **518**.

(41) In some aspects, methods, and systems for adjusting fluid injection into a wellbore based on relative permeability are provided according to one or more of the following examples:

(42) As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

(43) Example 1 is a computer implemented method comprising: determining actual trapped gas saturation data of a formation from maximum gas saturation data, maximum trapped gas saturation data, and actual gas saturation data; determining pseudo-maximum gas saturation data from the actual trapped gas saturation data, the maximum trapped gas saturation data, and the actual gas saturation data; determining relative permeability data of the formation by mapping the pseudo-maximum gas saturation data along a drainage curve; and adjusting fluid injection into the formation based on the relative permeability data.

(44) Example 2 is the method of example 1, wherein the maximum trapped gas saturation data and the actual gas saturation data are derived from analysis of a core plug extracted from the formation.

(45) Example 3 is the method of any of example(s) 1-2, wherein the pseudo-maximum gas saturation data is determined from the maximum trapped gas saturation data, the actual gas saturation data, and a synthetic value for the actual trapped gas saturation data in the following equation

(46)
$$S_{gmax_adj} = \frac{1}{((\frac{1}{S_{gtr}}) - (\frac{1}{S_{gr}}) + (\frac{1}{S_{gu}}))} \quad (\text{Eq. 2})$$

wherein the maximum trapped gas saturation is S_{gtr} , the actual gas saturation data is S_{gu} , the synthetic value for actual trapped gas saturation data is S_{gtr} , and the pseudo-maximum gas saturation data is S_{gmax_adj} .

(47) Example 4 is the method of any of example(s) 1-3, wherein the fluid injection is adjusted to alter a production rate of a water-alternating-gas hydrocarbon recovery process based on estimated total hydrocarbon reserves within the formation, derived from the relative permeability data.

(48) Example 5 is the method of any of example(s) 1-4, wherein the fluid injection is adjusted to alter a storage rate of a geological sequestration process in which carbon dioxide is stored in the formation as a supercritical fluid, based on an estimated capacity of the formation derived from the

relative permeability data as well as condensation effects derived from the actual trapped gas saturation data.

(49) Example 6 is the method of any of example(s) 1-5, wherein the actual trapped gas saturation data and the pseudo-maximum gas saturation data are consistent with the Land function.

(50) Example 7 is the method of any of the example(s) 1-6, wherein the pseudo-maximum gas saturation data is determined for subsequent values of the actual trapped gas saturation data from hysteresis modeling accounting for fluid entrapment from a non-wetting phase when the fluid entrapment from a wetting-phase is increased.

(51) Example 8 is a system comprising: a processor; and a non-transitory computer-readable medium including program code that is executable by the processor for causing the processor to perform operations comprising: determine actual trapped gas saturation data of a formation from maximum gas saturation data, maximum trapped gas saturation data, and actual gas saturation data; determine pseudo-maximum gas saturation data from the actual trapped gas saturation data, the maximum trapped gas saturation data, and the actual gas saturation data; determine relative permeability data of the formation by mapping the pseudo-maximum gas saturation data along a drainage curve; and adjust fluid injection into the formation based on the relative permeability data.

(52) Example 9 is the system of example 8, wherein the non-transitory computer-readable medium includes program code that is executable by the processor for causing the processor to perform operations comprising deriving the maximum trapped gas saturation data and the actual gas saturation data from analysis of a core plug extracted from the formation.

(53) Example 10 is the system of any of example(s) 8-9, wherein the non-transitory computer-readable medium includes program code that is executable by the processor for causing the processor to perform operations comprising determining the pseudo-maximum gas saturation data from the maximum trapped gas saturation data, the actual gas saturation data, and a synthetic value for the actual trapped gas saturation data in the following equation

$$(54) \quad S_{gmax_adj} = \frac{1}{((\frac{1}{S_{gtr}}) - (\frac{1}{S_{gr}}) + (\frac{1}{S_{gu}}))} \quad (\text{Eq. 2})$$

wherein the maximum trapped gas saturation is S_{gtr} , the actual gas saturation data is S_{gu} , the synthetic value for actual trapped gas saturation data is S_{gr} , and the pseudo-maximum gas saturation data is S_{gmax_adj} .

(55) Example 11 is the system of any of example(s) 8-10, wherein the non-transitory computer-readable medium includes program code that is executable by the processor for causing the processor to perform operations comprising adjusting the fluid injection to alter a production rate of a water-alternating-gas hydrocarbon recovery process based on estimated total hydrocarbon reserves within the formation, derived from the relative permeability data.

(56) Example 12 is the system of any of the example(s) 8-11, wherein the non-transitory computer-readable medium includes program code that is executable by the processor for causing the processor to perform operations comprising adjusting the fluid injection to alter a storage rate of a geological sequestration process, in which carbon dioxide is stored in the formation as a supercritical fluid, based on an estimated capacity of the formation derived from the relative permeability data as well as condensation effects derived from the actual trapped gas saturation data.

(57) Example 13 is the system of any of the example(s) 8-12, wherein the actual trapped gas saturation data and the pseudo-maximum gas saturation data are consistent with a Land function.

(58) Example 14 is the system of any of the example(s) 8-13, wherein non-transitory computer-readable medium includes program code that is executable by the processor for causing the processor to perform operations comprising determining the pseudo-maximum gas saturation data for subsequent values of the actual trapped gas saturation data from hysteresis modeling accounting for fluid entrapment from a non-wetting phase when the fluid entrapment from a wetting-phase is increased.

(59) Example 15 is a non-transitory computer-readable medium comprising program code that is executable by a processor for causing the processor to perform operations comprising: determine actual trapped gas saturation data of a formation from maximum gas saturation data, maximum trapped gas saturation data, and an actual gas saturation data; determine pseudo-maximum gas saturation data from the actual trapped gas saturation data, the maximum trapped gas saturation data, and the actual gas saturation data; determine relative permeability data of the formation by mapping the pseudo-maximum gas saturation data along a drainage curve; and adjust fluid injection into the formation based on the relative permeability data.

(60) Example 16 is the non-transitory computer-readable medium of example 15, wherein the operations further comprise deriving the maximum trapped gas saturation data and the actual gas saturation data from analysis of a core plug extracted from the formation.

(61) Example 17 is the non-transitory computer-readable medium of any of example(s) 15-16, wherein the operations further comprise determining the pseudo-maximum gas saturation data from the maximum trapped gas saturation data, the actual gas saturation data, and a synthetic value for the actual trapped gas saturation data in the following equation

$$(62) \quad S_{gmax_adj} = \frac{1}{((\frac{1}{S_{gtrap}}) - (\frac{1}{S_{gtr}}) + (\frac{1}{S_{gu}}))} \quad (Eq. 2)$$

wherein the maximum trapped gas saturation is S_{gtr} , the actual gas saturation data is S_{gu} , the synthetic value for actual trapped gas saturation data is S_{gtrap} , and the pseudo-maximum gas saturation data is S_{gmax_adj} .

(63) Example 18 is the non-transitory computer-readable medium of any of example(s) 15-17, wherein the operations further comprise adjusting the fluid injection to alter a production rate of a water-alternating-gas hydrocarbon recovery process based on estimated total hydrocarbon reserves within the formation, derived from the relative permeability data.

(64) Example 19 is the non-transitory computer readable medium of any of example(s) 15-18, wherein the operations further comprise adjusting the fluid injection to alter a storage rate of a geological sequestration process in which carbon dioxide is stored in the formation as a supercritical fluid, based on an estimated capacity of the formation derived from the relative permeability data as well as condensation effects derived from the actual trapped gas saturation data.

(65) Example 20 is the non-transitory computer readable medium of any of example(s) 15-19, wherein the actual trapped gas saturation data and the pseudo-maximum gas saturation data are consistent with a Land function.

(66) The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure.

Claims

1. A computer implemented method comprising: determining, by a processor of a computing device, actual trapped gas saturation data of a subsurface formation from maximum gas saturation data, maximum trapped gas saturation data, and actual gas saturation data; determining, by the processor of the computing device, pseudo-maximum gas saturation data from the actual trapped gas saturation data, the maximum trapped gas saturation data, and the actual gas saturation data; determining, by the processor of the computing device, relative permeability data of the subsurface formation by mapping the pseudo-maximum gas saturation data along a drainage curve; and affecting a hydrocarbon production rate of a hydrocarbon production well located in the subsurface formation by adjusting, by the processor of the computing device, fluid injection into an injection

well located in the subsurface formation based on the relative permeability data.

2. The method of claim 1, wherein the maximum trapped gas saturation data and the actual gas saturation data are derived from analysis of a core plug extracted from the subsurface formation.

3. The method of claim 1, wherein the pseudo-maximum gas saturation data is determined from the maximum trapped gas saturation data, the actual gas saturation data, and a synthetic value for the actual trapped gas saturation data in the following equation $S_{gmax_adj} = \frac{1}{((\frac{1}{S_{gtrap}}) - (\frac{1}{S_{gr}}) + (\frac{1}{S_{gu}}))}$ wherein the maximum trapped gas saturation is S_{gr} , the actual gas saturation data is S_{gu} , the synthetic value for actual trapped gas saturation data is S_{gtrap} , and the pseudo-maximum gas saturation data is S_{gmax_adj} .

4. The method of claim 1, wherein the fluid injection is adjusted by the computing device to affect the hydrocarbon production rate of a water-alternating-gas hydrocarbon recovery process based on estimated total hydrocarbon reserves within the subsurface formation, derived from the relative permeability data.

5. The method of claim 1, wherein the fluid injection is adjusted by the computing device to alter a storage rate of a geological sequestration process in which carbon dioxide is stored in the subsurface formation as a supercritical fluid, based on an estimated capacity of the subsurface formation derived from the relative permeability data as well as condensation effects derived from the actual trapped gas saturation data.

6. The method of claim 1, wherein the actual trapped gas saturation data and the pseudo-maximum gas saturation data are consistent with a Land function.

7. The method of claim 6, wherein the pseudo-maximum gas saturation data is determined for subsequent values of the actual trapped gas saturation data from hysteresis modeling accounting for fluid entrapment from a non-wetting phase when the fluid entrapment from a wetting-phase is increased.

8. A system comprising: a computing device including a processor and a non-transitory computer-readable medium including program code that is executable by the processor for causing the computing device to perform operations comprising: determining actual trapped gas saturation data of a subsurface formation from maximum gas saturation data, maximum trapped gas saturation data, and actual gas saturation data, determining pseudo-maximum gas saturation data from the actual trapped gas saturation data, the maximum trapped gas saturation data, and the actual gas saturation data, and determining relative permeability data of the subsurface formation by mapping the pseudo-maximum gas saturation data along a drainage curve; and wherein the computing device is communicatively couplable to an injection distribution system associated with an injection well located in the subsurface formation to affect a hydrocarbon production rate of a hydrocarbon production well located in the subsurface formation by adjusting fluid injection from the injection distribution system into the injection well based on the relative permeability data.

9. The system of claim 8, wherein the operations further comprise deriving the maximum trapped gas saturation data and the actual gas saturation data from analysis of a core plug extracted from the subsurface formation.

10. The system of claim 8, wherein the non-transitory computer-readable medium includes program code that is executable by the processor for causing the processor to perform operations comprising determining the pseudo-maximum gas saturation data from the maximum trapped gas saturation data, the actual gas saturation data, and a synthetic value for the actual trapped gas saturation data in the following equation $S_{gmax_adj} = \frac{1}{((\frac{1}{S_{gtrap}}) - (\frac{1}{S_{gr}}) + (\frac{1}{S_{gu}}))}$ wherein the maximum trapped gas saturation is S_{gr} , the actual gas saturation data is S_{gu} , the synthetic value for actual trapped gas saturation data is S_{gtrap} , and the pseudo-maximum gas saturation data is S_{gmax_adj} .

11. The system of claim 8, wherein the fluid injection is part of a water-alternating-gas hydrocarbon recovery process and the operations further comprise adjusting the fluid injection based on estimated total hydrocarbon reserves within the subsurface formation, derived from the

relative permeability data.

12. The system of claim 8, wherein the operations further comprise adjusting the fluid injection to alter a storage rate of a geological sequestration process, in which carbon dioxide is stored in the subsurface formation as a supercritical fluid, based on an estimated capacity of the substrate formation derived from the relative permeability data as well as condensation effects derived from the actual trapped gas saturation data.

13. The system of claim 8, wherein the actual trapped gas saturation data and the pseudo-maximum gas saturation data are consistent with a Land function.

14. The system of claim 13, wherein the operations further comprising determining the pseudo-maximum gas saturation data for subsequent values of the actual trapped gas saturation data from hysteresis modeling accounting for fluid entrapment from a non-wetting phase when the fluid entrapment from a wetting-phase is increased.

15. A non-transitory computer-readable medium comprising program code that is executable by a processor of a computing device for causing the processor to perform operations comprising: determining actual trapped gas saturation data of a subsurface formation from maximum gas saturation data, maximum trapped gas saturation data, and an actual gas saturation data; determining pseudo-maximum gas saturation data from the actual trapped gas saturation data, the maximum trapped gas saturation data, and the actual gas saturation data; determining relative permeability data of the formation by mapping the pseudo-maximum gas saturation data along a drainage curve; and affecting a hydrocarbon production rate of a hydrocarbon production well located in the subsurface formation by adjusting fluid injection into an injection well located in the subsurface formation based on the relative permeability data.

16. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise deriving the maximum trapped gas saturation data and the actual gas saturation data from analysis of a core plug extracted from the subsurface formation.

17. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise determining the pseudo-maximum gas saturation data from the maximum trapped gas saturation data, the actual gas saturation data, and a synthetic value for the actual trapped gas saturation data in the following equation $S_{\text{gmax_adj}} = \frac{1}{((\frac{1}{S_{\text{gtrap}}}) - (\frac{1}{S_{\text{gtr}}}) + (\frac{1}{S_{\text{gu}}}))}$ wherein the maximum

trapped gas saturation is S_{gtr} , the actual gas saturation data is S_{gu} , the synthetic value for actual trapped gas saturation data is S_{gtrap} , and the pseudo-maximum gas saturation data is $S_{\text{gmax_adj}}$.

18. The non-transitory computer-readable medium of claim 15, wherein the fluid injection is part of a water-alternating-gas hydrocarbon recovery process and the operations further comprise adjusting the fluid injection based on estimated total hydrocarbon reserves within the subsurface formation, derived from the relative permeability data.

19. The non-transitory computer-readable medium of claim 15, wherein the operations further comprise adjusting the fluid injection to alter a storage rate of a geological sequestration process in which carbon dioxide is stored in the subsurface formation as a supercritical fluid, based on an estimated capacity of the subsurface formation derived from the relative permeability data as well as condensation effects derived from the actual trapped gas saturation data.

20. The non-transitory computer-readable medium of claim 15, wherein the actual trapped gas saturation data and the pseudo-maximum gas saturation data are consistent with a Land function.
